

Q&A — AS 7 — Construction Contracts

Companion question bank to the concept chapter. All amounts in Indian Rupees (₹). Based on **AS 7 (Revised 2002)** as notified by ICAI. The revised standard permits **only the Percentage of Completion Method (POCM)**; the erstwhile Completed Contract Method stands withdrawn.

Section A — Concept-Check Questions (test the WHY)

A1. Why does AS 7 mandate the Percentage of Completion Method and not the Completed Contract Method? Construction contracts often span more than one accounting period. If revenue were recognised only on completion, the P&L of intermediate years would show cost with no matching revenue, distorting periodic performance and violating the accrual and matching concepts. POCM spreads revenue and profit over the periods in which the contract activity (and hence the value addition) actually occurs, giving a truer periodic picture.

A2. What is the single most important asymmetry in AS 7 between profits and losses? Profit is recognised **gradually** by reference to the stage of completion. But an **expected loss on the whole contract is recognised immediately and in full** as an expense, irrespective of the stage of completion or whether work has commenced. This is the application of prudence/conservatism.

A3. When the outcome of a contract cannot be estimated reliably, how is revenue recognised? Revenue is recognised **only to the extent of contract costs incurred that it is probable will be recoverable**, and contract costs are recognised as an expense in the period incurred. The result is **nil profit** — this is often called the "cost recovery" or "zero-profit" approach. Once the uncertainty is removed, normal POCM resumes.

A4. Give two examples of the "outcome cannot be estimated reliably" situation. (i) A fixed-price contract where total contract costs cannot be reliably estimated; (ii) A contract whose enforceability is in serious doubt (e.g., litigation, pending regulatory approval), so recovery of costs is uncertain.

A5. Why are costs relating to future activity excluded when measuring stage of completion under the cost-to-cost method? Stage of completion must reflect work actually performed. Materials delivered to site but not yet installed, or advance payments to sub-contractors for work not yet done, represent future activity. Including them would overstate the percentage complete and prematurely recognise profit. They are excluded from "costs incurred to date" for the ratio (but shown as WIP/asset).

A6. What is the difference between "contract revenue" recognised and "progress billings"? Contract revenue is recognised on an **earned** basis using stage of completion, independent of amounts invoiced. Progress billings are amounts **invoiced** to the customer per the contract's payment schedule. The two rarely match in a period; the difference (with costs and profits) creates the "due from / due to customer" balance-sheet item.

A7. How are variations, claims and incentive payments treated in contract revenue? They are included **only when** it is **probable** they will result in revenue **and** they are **capable of being reliably measured**. Claims carry extra uncertainty (negotiation/legal outcome), so the threshold is applied strictly.

A8. Distinguish a fixed-price contract from a cost-plus contract for the purpose of "reliable estimation". For a **fixed-price** contract, outcome is reliably estimable when total revenue and total costs can be reliably measured, it is probable economic benefits will flow, and costs to complete plus stage of completion can be measured. For a **cost-plus** contract, the test is lighter — mainly that it is probable benefits will flow and contract costs (whether or not reimbursable) can be clearly identified and measured.

Section B — Graded Computational Problems

B1 (Easy) — Basic cost-to-cost POCM

A contractor has a fixed-price contract for ₹100 lakh. Costs incurred to date ₹40 lakh; estimated further costs ₹40 lakh. Compute stage of completion, and revenue and profit to recognise.

Solution. Total estimated cost = $40 + 40 = ₹80$ lakh. Stage of completion = $40 / 80 = 50\%$. Total expected profit = $100 - 80 = ₹20$ lakh. Revenue recognised = $50\% \times 100 = ₹50$ lakh. Cost recognised = ₹40 lakh (actual to date). Profit = $50 - 40 = ₹10$ lakh ($= 50\% \times 20$). ✓

B2 (Easy–Moderate) — Excluding future-activity costs

Contract price ₹500 lakh. Costs incurred to date ₹390 lakh, which **includes ₹90 lakh of materials lying at site not yet used**. Estimated further costs ₹110 lakh. Determine profit to date.

Solution. Costs of **work performed** = $390 - 90 = ₹300$ lakh (materials not used relate to future activity). Total estimated cost = costs of work performed 300 + unused material 90 + further costs 110 = ₹500 lakh? Careful: total estimated cost = actual costs incurred (390) + further costs to complete (110) = ₹500 lakh. Stage of completion (cost-to-cost) = costs of work performed ÷ total estimated cost = $300 / 500 = 60\%$. Total profit = $500 - 500 = \text{Nil}$.

Recompute with a genuine margin. Say contract price is ₹600 lakh (correction for illustration): total profit = $600 - 500 = ₹100$ lakh; revenue = $60\% \times 600 = ₹360$ lakh; cost recognised = ₹300 lakh; profit = ₹60 lakh. The ₹90 lakh unused material is carried as an **asset (WIP)**, not expensed, and excluded from the percentage.

Trap: including the ₹90 lakh in the numerator would give $390/500 = 78\%$ and overstate profit.

B3 (Moderate) — Expected loss recognised in full

Contract price ₹120 lakh. Costs incurred to date ₹80 lakh; estimated further costs ₹60 lakh. Stage of completion by surveys = 60%. Determine the amounts to be taken to P&L.

Solution. Total estimated cost = $80 + 60 = ₹140$ lakh. Total expected result = $120 - 140 = \text{loss of ₹20 lakh}$. Under AS 7, an expected loss is recognised **immediately and in full**, regardless of stage of completion.

Revenue recognised = $60\% \times 120 = ₹72$ lakh. Cost to be charged so that P&L shows the full ₹20 lakh loss: Cost charged = Revenue recognised + Total expected loss = $72 + 20 = ₹92$ lakh. P&L: Revenue 72 – Cost 92 = **(₹20 lakh) loss**. ✓ (Equivalently: profit as per stage = $60\% \times (-20) = -12$; add the *additional* provision of -8 to reach the full -20 .)

B4 (Moderate) — "Due from / due to customer" presentation

A contract shows: costs incurred to date ₹250 lakh; recognised profits ₹40 lakh; recognised losses Nil; progress billings ₹260 lakh; advances received ₹30 lakh; retention by customer ₹15 lakh. Show the balance-sheet item.

Solution. Gross amount **due from / (due to)** customers = Costs incurred + Recognised profits – Recognised losses – Progress billings = $250 + 40 - 0 - 260 = ₹30$ lakh (**due FROM customer** → **shown as an ASSET**).

Presentation: - Gross amount due from customers (asset) = ₹30 lakh. - Advances received ₹30 lakh → shown as a **liability** (not netted). - Retention ₹15 lakh → part of receivables/amounts due; disclosed separately.

If the figure had been negative it would be shown as **gross amount due to customers (a liability)**.

B5 (Exam-hard) — Three-year contract with a mid-way revision

A contractor undertakes a 3-year fixed-price contract for **₹300 lakh**. Data:

Particulars	Year 1	Year 2	Year 3
Costs incurred to date	90	210	305
Estimated further costs	180	90	—
Progress billings (cumulative)	100	200	300

Compute revenue, cost and profit for each year using the cost-to-cost method.

Solution.

Year 1 Total estimated cost = $90 + 180 = ₹270$ lakh. Total profit = $300 - 270 = ₹30$ lakh. Stage = $90/270 = 33.33\%$. Revenue = $33.33\% \times 300 = ₹100$ lakh. Cost = ₹90 lakh. **Profit Y1 = ₹10 lakh.**

Year 2 Total estimated cost = 210 + 90 = ₹300 lakh. Total profit = 300 – 300 = **Nil**. Stage = 210/300 = 70%. Cumulative revenue = 70% × 300 = ₹210 lakh → Revenue Y2 = 210 – 100 = ₹110 lakh. Cumulative profit recognised should now be Nil, but ₹10 lakh was recognised in Y1. Cost Y2 (balancing) = Revenue Y2 – required profit change = 110 – (0 – 10) = ₹120 lakh (= 210 – 90 actual costs). **Profit Y2 = 110 – 120 = (₹10 lakh)** — reverses Y1's profit; cumulative profit = Nil. ✓

Year 3 Contract complete. Actual total cost = ₹305 lakh. Total result = 300 – 305 = **loss ₹5 lakh**. Cumulative revenue = ₹300 lakh → Revenue Y3 = 300 – 210 = ₹90 lakh. Cost Y3 = 305 – 210 = ₹95 lakh. **Profit Y3 = 90 – 95 = (₹5 lakh)**.

Summary: Y1 +10, Y2 –10, Y3 –5 → net **loss ₹5 lakh** = actual outcome. ✓ *Examiner point:* the moment Year 2 estimates showed no profit, prior profit was corrected as a **change in estimate** (prospectively, through current-year P&L — no restatement of Year 1).

B6 (Exam-hard) — Journal entries for one year

Using **Year 1 of B5** (Revenue ₹100 lakh, Cost ₹90 lakh, billings ₹100 lakh, cash received ₹80 lakh), pass journal entries.

Solution (₹ in lakh).

1. Contract costs incurred:

Work-in-Progress (Contract) A/c	Dr	90	
To Cash/Creditors A/c			90

2. Progress bill raised on customer:

Contractee (Customer) A/c	Dr	100	
To Contract WIP / Billings A/c			100

3. Cash received from customer:

Bank A/c	Dr	80	
To Contractee A/c			80

4. Revenue & cost recognition (P&L):

Cost of Construction (P&L) A/c	Dr	90	
Contract WIP A/c	Dr	10	(profit loaded to WIP)
To Contract Revenue (P&L) A/c			100

Balance sheet: Amount due from customer = Costs 90 + Profit 10 – Billings 100 = **Nil**; Contractee shows ₹20 lakh still receivable (100 billed – 80 received).

Section C — Past-Paper-Style Full Questions

C1. (ICAI pattern — reliable vs unreliable outcome)

SunTech Ltd. began a contract on 1 April for ₹350 lakh. By 31 March costs incurred were ₹120 lakh. Owing to a boundary-title dispute with the customer, the contractor is uncertain whether the contract is enforceable, though it is probable the ₹120 lakh already spent will be recovered. Estimated further costs cannot be reliably measured. How should SunTech recognise revenue and costs?

Model answer. The outcome of the contract **cannot be estimated reliably** because total costs to complete cannot be measured and enforceability is doubtful. Under AS 7, in such a case: - Revenue is recognised **only to the extent of contract costs incurred that are probable of recovery = ₹120 lakh**. - Contract costs are recognised as an expense in the period incurred = **₹120 lakh**. - **Profit recognised = Nil**. If, in a later year, the dispute is resolved and costs to complete become reliably estimable, SunTech switches to normal POCM prospectively. Had recovery of even the ₹120 lakh been improbable, revenue would be Nil and the full ₹120 lakh charged as expense (an expected loss).

C2. (ICAI pattern — full computation with disclosure)

Bharat Infra Ltd. has a contract: total price ₹900 lakh. At year-end — costs incurred to date ₹520 lakh (including ₹40 lakh of standard materials at site not yet consumed and ₹20 lakh advance to a sub-contractor for future work); estimated further costs ₹300 lakh; progress billings ₹480 lakh; amounts received ₹400 lakh. Compute profit, and show amounts for the balance sheet and disclosures.

Model answer. Costs relating to **future activity** to be excluded from stage of completion = $40 + 20 = ₹60$ lakh. Costs of work performed = $520 - 60 = ₹460$ lakh. Total estimated cost = $520 + 300 = ₹820$ lakh. Total expected profit = $900 - 820 = ₹80$ lakh. Stage of completion = $460 / 820 = 56.10\%$. Revenue recognised = $56.10\% \times 900 = ₹504.9$ lakh. Cost recognised (of work performed) = ₹460 lakh. **Profit recognised = $504.9 - 460 = ₹44.9$ lakh** ($\approx 56.10\% \times 80$).

Balance sheet — Gross amount due from customers = Costs incurred (of work performed) + Recognised profit – Progress billings = $460 + 44.9 - 480 = ₹24.9$ lakh (**asset**). The ₹60 lakh future-activity cost is carried separately as **inventory / WIP (asset)**.

Disclosures (AS 7 para 38–42): - Contract revenue recognised in the period: ₹504.9 lakh. - Method of revenue recognition: percentage of completion. - Method of determining stage of completion: proportion of contract costs incurred (cost-to-cost). - Aggregate costs incurred and recognised profits (net of losses) to date: $520 + 44.9 = ₹564.9$ lakh. - Progress billings ₹480 lakh; amounts received ₹400 lakh; customer still owes $480 - 400 = ₹80$ lakh.

C3. (ICAI pattern — foreseeable loss + prior profit)

Vega Constructions had recognised ₹18 lakh profit on a contract up to last year. This year, cost escalation has made the contract loss-making: total price ₹200 lakh, total estimated cost now ₹230 lakh, costs to date ₹138 lakh. What should this year's P&L show?

Model answer. Total expected result = 200 – 230 = **loss ₹30 lakh**. AS 7 requires the **entire expected loss ₹30 lakh** to be recognised **immediately**. Since ₹18 lakh profit was recognised earlier, the current year must both **reverse that ₹18 lakh** and **book the ₹30 lakh loss**: Amount to be charged this year = 30 + 18 = **₹48 lakh loss**. This is treated as a **change in accounting estimate**, adjusted prospectively in the current year's P&L; prior-period figures are **not** restated. The excess of expected loss over costs incurred is, in substance, a **provision for foreseeable losses**.

Section D — MCQs / Case Scenarios (30% MCQ pattern)

D1. Under revised AS 7, the permitted method of accounting for construction contracts is: (a) Completed contract (b) Percentage of completion (c) Either (a) or (b) (d) Cash basis **Ans: (b).** Revised AS 7 (2002) allows only POCM; completed-contract method was withdrawn.

D2. An expected loss on a contract is recognised: (a) Over the remaining life (b) Only on completion (c) Immediately, in full (d) In proportion to stage **Ans: (c).** Prudence — the whole foreseeable loss is charged at once.

D3. Costs incurred ₹60 lakh, further costs ₹60 lakh, price ₹150 lakh. Profit recognised to date: (a) ₹15 lakh (b) ₹30 lakh (c) ₹75 lakh (d) ₹150 lakh **Ans: (a).** Stage 60/120 = 50%; total profit 30; recognised 50% × 30 = ₹15 lakh.

D4. When outcome cannot be estimated reliably, profit recognised is: (a) Full (b) Based on stage (c) Nil (d) 50% **Ans: (c).** Revenue = recoverable costs; costs = expense; profit is Nil.

D5. Materials delivered to site but not yet used, under cost-to-cost method, are: (a) Included in costs incurred for stage % (b) Excluded — they relate to future activity (c) Expensed immediately (d) Added to revenue **Ans: (b).** They represent future activity; carried as asset, excluded from the ratio.

D6. "Gross amount due to customers" equals costs + recognised profits – losses – progress billings, and is shown as: (a) An asset (b) A liability (c) Revenue (d) Equity **Ans: (b).** A negative figure (billings exceed costs + profit) is a liability.

D7. Claims and variations are included in contract revenue only when: (a) Billed (b) Received in cash (c) Probable and reliably measurable (d) Approved by auditor **Ans: (c).** Both conditions — probability of revenue and reliable measurement.

D8. Contract revenue ₹80 lakh recognised, but only ₹65 lakh billed. The ₹15 lakh difference (with costs) forms part of: (a) Advances (b) Retention money (c) Amount due from customer (asset) (d) Deferred revenue **Ans: (c).** Earned-but-unbilled portion sits in "due from customer".

D9. Advances received from a customer are: (a) Netted against WIP (b) Shown as a liability separately (c) Revenue (d) Deducted from costs **Ans: (b).** Disclosed as a liability, not netted against the due-from-customer asset.

D10. A cost-plus contract's outcome is reliably estimable mainly when: (a) Fixed price known (b) It is probable benefits will flow and costs are identifiable/measurable (c) 50% complete (d) Fully billed **Ans: (b).** Cost-plus needs a lighter test than fixed-price.

D11 (Case). *Contract price ₹400 lakh; costs to date ₹300 lakh; further costs ₹150 lakh; stage by survey 65%.* The amount charged to P&L as cost this year (so full loss shows) is: (a) ₹300 lakh (b) ₹310 lakh (c) ₹260 lakh (d) ₹315 lakh **Ans: (b).** Total cost 450 → loss ₹50 lakh. Revenue = $65\% \times 400 = ₹260$ lakh. Cost = revenue + loss = $260 + 50 = ₹310$ lakh.

D12. Change in estimated total contract cost is accounted for as: (a) Prior-period item (b) Change in accounting policy (c) Change in accounting estimate — prospectively (d) Extraordinary item **Ans: (c).** AS 7 estimate revisions flow through the current period's P&L.

End of Q&A — AS 7 Construction Contracts.